

Kevin Lin

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Education

University of Waterloo

Sept 2024 – May 2029

BASc in Systems Design Engineering

CGPA: 93.0/100 (4.0)

Work Experience

Software Engineering Intern - Rocket Companies

May 2026 - Present

- Owned end-to-end delivery of native in-app feedback collection for Android, scoping, building, and launching the module independently from spec to production on **Google Play** for a **top-10 Finance app**.
- Replaced a WebView-based CSAT survey with a native **Jetpack Compose** module (**MVVM, Hilt**), raising expected response rates from **1%** toward the **13%** industry average for native surveys.

Software Engineering Intern - Adapt Media

Sept 2025 - Dec 2025

- Optimized parent-child communication by implementing a **multithreaded Java long-polling server** with lock-based coordination to ensure race-free command delivery, decreasing latency by **40%**.
- Built an offline pairing protocol using **QR-based** key exchange and an **ECDH & HKDF** derived shared secret to mutually authenticate devices, completing pairing in **<5s**.

Machine Learning Intern - Government of Canada

Apr 2025 - Aug 2025

- Benchmarked four models (**Logistic Regression, Gradient Boosting, SVM, Random Forest**) on electricity demand forecasting across **6 regions**, selecting the best-fit model per region to support grid load balancing.
- Reduced forecast **RMSE by 20%** by resampling and normalizing time-series usage data prior to training, stabilizing accuracy across regions.
- Evaluated forecast accuracy using **MAE, RMSE, and R²** per region to justify model selection.

Projects

🔗 MindView | Python, TypeScript, Google Gemini, MongoDB, CNN, Scikit-Image, NiBabel 2026

- Built MindView, a clinician-facing web app that turns fragmented 2D MRI slices into an interactive, **color-coded 3D brain + tumor visualization**, reducing manual cross-slice comparison.
- Fine-tuned **MONAI SegResNet (3D CNN)** to segment tumor subregions into **voxel grids** from MRI volumes, then generated controllable **3D meshes** (Nibabel/scikit-image) with opacity + segment controls.
- Finalist** at DeltaHacks 2026 from over 700 participants.

Semantic Search & Recommendation System | Python, OpenAI Embeddings, FAISS, ChromaDB 2025

- Built a **semantic search** pipeline over a movie corpus using batched **OpenAI embeddings**, validating cluster quality via **cosine similarity** before tuning.
- Benchmarked an **approximate nearest-neighbor index** against exact search on up to **50,000 vectors**, achieving a **7.1x** query speedup (5.81ms → 0.82ms).
- Built precision/recall evaluation from scratch to compare embedding models, finding a **28%** precision and **46%** recall improvement at top-5 results.

🔗 Search Engine Autocomplete | C++, Python, FastAPI, React, Redis, Git 2025

- Built a **trie-based** autocomplete search engine with ranked phrase suggestions.
- Built a responsive, real time search bar UI using React and TailwindCSS, achieving **sub-100 ms** updates.
- Implemented an **LRU cache** on Redis for frequent search prefixes, reducing average query latency by **90%**.

Technical Skills

Languages: Python, Java, C++, Kotlin, JavaScript, TypeScript, SQL, HTML/CSS

Frameworks/Libraries: React, Node.js, Express.js, Tailwind CSS, FastAPI, Pandas, Scikit-learn, Matplotlib, Zustand

Tools: PostgreSQL, MongoDB, Android Studio, Git, Redis, VSCode, Jupyter Notebooks, Gradle, Prisma